

Homework 1

Electromagnetism - PHYS 715

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1.1 Use Gauss's theorem [and (1.21) if necessary] to prove the following:

- (a) Any excess charge placed on a conductor must lie entirely on its surface. (A conductor by definition contains charges capable of moving freely under the action of applied electric fields.)
- (b) A closed, hollow conductor shields its interior from fields due to charges outside, but does not shield its exterior from the fields due to charges placed inside it.
- (c) The electric field at the surface of a conductor is normal to the surface and has a magnitude σ/ϵ_0 , where σ is the charge density per unit area on the surface.

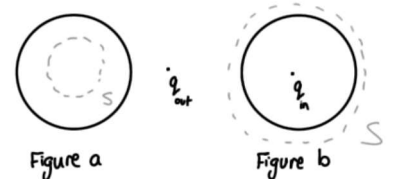
1.1 a) A conductor has $\vec{E} = 0$ inside its volume. Suppose $\exists$ a charge inside the volume then it will have a non-zero electric field inside. $\therefore$ The only place for extra charge to go is its surface. $\vec{E} = 0, \vec{\nabla} \cdot \vec{E} = 0 = \rho/\epsilon_0 \therefore \rho = 0$

b) Drawing a Gaussian surface inside the conductor (Figure a)
The electric field inside it must be zero. (No charge in S)
Therefore, the induced charge on the conductor counteracts the charge outside.

Drawing a Gaussian Surface outside the conductor (Figure b)

$$\int_S \vec{E} \cdot d\vec{a} = \frac{q_{in}}{\epsilon_0} \quad \vec{E}(r) = \frac{1}{4\pi\epsilon_0} \frac{q_{in}}{r^2} \hat{r}$$

$\therefore \exists$ an electric field outside the conductor



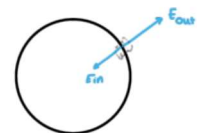
c) We consider a tiny Gaussian pill box on the surface of the conductor.

$$\int_S \vec{E} \cdot \hat{n} da = \frac{q_{in}}{\epsilon_0}$$

$$(\vec{E}_{out} - \vec{E}_{in}) \cdot \hat{n} \cdot A = \frac{\sigma A}{\epsilon_0}$$

conductor

$$\vec{E}_{out} \cdot \hat{n} = \frac{\sigma}{\epsilon_0}$$



Suppose that $\vec{E}_{out}$ has a tangential component,

then charge on the surface would experience a force allowing it to move on the surface.

But, this violates the assumption of electrostatics!

$$\therefore \vec{E}_{out} \parallel \hat{n}$$

1.3 Using Dirac delta functions in the appropriate coordinates, express the following charge distributions as three-dimensional charge densities $\rho(\mathbf{x})$.

- (a) In spherical coordinates, a charge Q uniformly distributed over a spherical shell of radius R .
- (b) In cylindrical coordinates, a charge λ per unit length uniformly distributed over a cylindrical surface of radius b .
- (c) In cylindrical coordinates, a charge Q spread uniformly over a flat circular disc of negligible thickness and radius R .
- (d) The same as part (c), but using spherical coordinates.

$$a) \rho(r, \theta, \phi) = \frac{Q}{4\pi R^2} \delta(r-R)$$

$$b) \rho(s, \phi, z) = \frac{\lambda}{2\pi b} \delta(s-b)$$

$$c) \rho(s, \phi, z) = \begin{cases} \frac{Q}{\pi R^2} \delta(z) & \text{for } s \leq R \\ 0 & \text{for } s > R \end{cases} \quad \text{disc is on } z=0$$

$$d) \rho(r, \theta, \phi) = \begin{cases} \frac{Q}{\pi R^2} \frac{\delta(\theta - \pi/2)}{r} & \text{for } r \leq R \\ 0 & \text{for } r > R \end{cases} \quad \text{disc is on } \theta = \pi/2$$

Some notes to discuss:

$$* \rho = \frac{Q}{A} \quad \text{for the region of interest.}$$

* For part d, we obtain an extra $\frac{1}{r}$ factor due to how delta functions work:

$$z = r \cos \theta$$

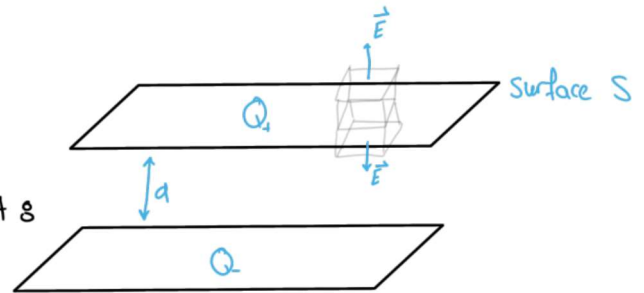
$$\delta(f(x)) = \sum_i \frac{\delta(x-x_i)}{|f'(x_i)|}$$

$$\delta(z) = \frac{\delta(\theta - \pi/2)}{|r \sin \pi/2|} = \frac{\delta(\theta - \pi/2)}{r}$$

- 1.6 A simple capacitor is a device formed by two insulated conductors adjacent to each other. If equal and opposite charges are placed on the conductors, there will be a certain difference of potential between them. The ratio of the magnitude of the charge on one conductor to the magnitude of the potential difference is called the capacitance (in SI units it is measured in farads). Using Gauss's law, calculate the capacitance of
- two large, flat, conducting sheets of area A , separated by a small distance d ;
 - two concentric conducting spheres with radii a, b ($b > a$);
 - two concentric conducting cylinders of length L , large compared to their radii a, b ($b > a$).
 - What is the inner diameter of the outer conductor in an air-filled coaxial cable whose center conductor is a cylindrical wire of diameter 1 mm and whose capacitance is $3 \times 10^{-11} \text{ F/m}$? $3 \times 10^{-12} \text{ F/m}$?

1.6

a) We attempt to find the electric field E first



$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$E(2A) = \frac{Q}{\epsilon_0}$$

$E = \frac{Q}{2A\epsilon_0}$ from one plate. Electric field between both plates is $E = 2 \frac{Q}{2A\epsilon_0} = \frac{Q}{A\epsilon_0}$

$$V = \int E \, dl = E \int dl = Ed = \frac{Qd}{A\epsilon_0}$$

$$C = \frac{Q}{V} = \frac{A\epsilon_0}{d}$$

b) The electric field from a conducting sphere is

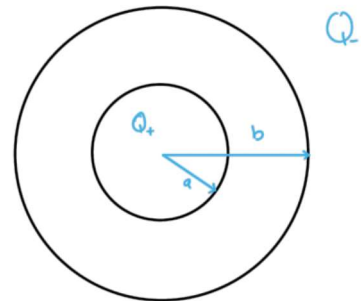
$$\vec{E}(\vec{r}) = E(r) \hat{r}$$

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$\int E(r) \hat{r} \cdot \hat{n} \, dA = \frac{Q}{\epsilon_0}$$

$$E(r) (4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad \text{or} \quad \vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$



$$V = \int_a^b \vec{E}(\vec{r}) \cdot d\vec{l} = \int_a^b \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \, dr = -\frac{1}{4\pi\epsilon_0} \left(\frac{Q}{b} - \frac{Q}{a} \right) = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$C = \frac{Q}{V} = 4\pi\epsilon_0 \left(\frac{ab}{b-a} \right)$$

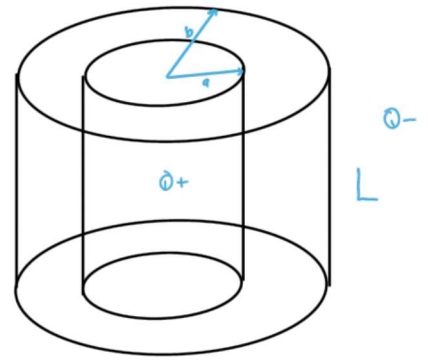
$$c) \quad E(2\pi rL) = \frac{Q}{\epsilon_0}$$

$$\vec{E} = \frac{Q}{2\pi rL\epsilon_0} \hat{r}$$

$$\int_a^b \vec{E} \cdot d\vec{\ell} = \int_a^b \frac{Q}{2\pi r\epsilon_0 L} dr$$

$$= \frac{Q}{2\pi\epsilon_0 L} (\ln b - \ln a)$$

$$C = \frac{Q}{V} = \frac{2\pi\epsilon_0 L}{\ln(b/a)}$$



d) This problem requires us to solve for radius b . (diameter b)

$$\ln\left(\frac{b}{a}\right) = \frac{2\pi\epsilon_0 L}{C}$$

$$\Rightarrow b = ae^{\frac{2\pi\epsilon_0 L}{C}} \quad \text{OR} \quad b = ae^{\frac{2\pi\epsilon_0}{C/L}}$$

$$\text{For } C/L = 3 \times 10^{-11} \text{ F/m,}$$

$$b = (10^{-3} \text{ m}) e^{\frac{2\pi(8.85 \times 10^{-12}) \text{ F/m}}{3 \times 10^{-11} \text{ F/m}}}$$

$$\approx 6.36 \text{ mm}$$

$$\text{For } C/L = 3 \times 10^{-12} \text{ F/m,}$$

$$b = (10^{-3} \text{ m}) e^{\frac{2\pi(8.85 \times 10^{-12}) \text{ F/m}}{3 \times 10^{-12} \text{ F/m}}}$$

$$\approx 101 \text{ km}$$

1.5 The time-averaged potential of a neutral hydrogen atom is given by

$$\Phi = \frac{q}{4\pi\epsilon_0} \frac{e^{-\alpha r}}{r} \left(1 + \frac{\alpha r}{2} \right)$$

where q is the magnitude of the electronic charge, and $\alpha^{-1} = a_0/2$, a_0 being the Bohr radius. Find the distribution of charge (both continuous and discrete) that will give this potential and interpret your result physically.

1.5

We would like to find $\rho(r)$ that would give Φ presented here.

A couple of notes are in order.

- 1) $\nabla^2 \Phi = -\rho(\vec{r})/\epsilon_0$
- 2) We will use spherical coordinates of the laplacian as we have a spherically symmetric system
- 3) We will need to be wary of the proton. (charge = q but at $r=0$)

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right)$$

$$\frac{d\Phi}{dr} = \frac{q}{4\pi\epsilon_0} \left[\frac{-\alpha r e^{-\alpha r} - e^{-\alpha r}}{r^2} \left(1 + \frac{\alpha r}{2} \right) + \frac{e^{-\alpha r}}{r} \left(\frac{\alpha}{2} \right) \right]$$

$$\begin{aligned} r^2 \frac{d\Phi}{dr} &= \frac{q}{4\pi\epsilon_0} \left[(-\alpha r e^{-\alpha r} - e^{-\alpha r}) \left(1 + \frac{\alpha r}{2} \right) + r e^{-\alpha r} \left(\frac{\alpha}{2} \right) \right] \\ &= \frac{q}{4\pi\epsilon_0} e^{-\alpha r} \left[-\alpha r - \frac{\alpha^2 r^2}{2} - 1 \right] \\ &= \frac{-q}{4\pi\epsilon_0} e^{-\alpha r} \left[1 + \alpha r + \frac{\alpha^2 r^2}{2} \right] \end{aligned}$$

$$\begin{aligned} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) &= \frac{-q}{4\pi\epsilon_0} \left[e^{-\alpha r} (\alpha + \alpha^2 r) - \alpha e^{-\alpha r} \left(1 + \alpha r + \frac{\alpha^2 r^2}{2} \right) \right] \\ &= \frac{-q}{4\pi\epsilon_0} e^{-\alpha r} \left(-\frac{\alpha^3 r^2}{2} \right) \end{aligned}$$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = \frac{q\alpha^3}{8\pi\epsilon_0} e^{-\alpha r} \Rightarrow \rho(r) = -\nabla^2 \Phi \epsilon_0 = \frac{-q\alpha^3}{8\pi} e^{-\alpha r}$$

$$\rho(r) = q \delta^3(\vec{r})$$

$$\begin{aligned} \text{since } \nabla^2 \Phi &= \nabla^2 \left(\frac{q}{4\pi\epsilon_0} \frac{1}{r} \right) \\ &= \frac{q}{4\pi\epsilon_0} (-4\pi \delta^3(\vec{r})) \\ &= -\frac{q}{\epsilon_0} \delta^3(\vec{r}) = -\frac{\rho(\vec{r})}{\epsilon_0} \\ \Rightarrow \rho(\vec{r}) &= q \delta^3(\vec{r}) \end{aligned}$$

Therefore, our charge distribution $\rho(r)$ is

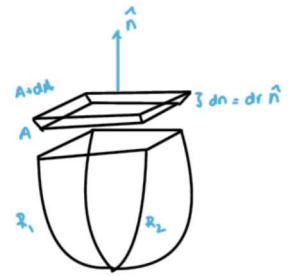
$$\rho(r) = q \delta^3(\vec{r}) - \frac{q\alpha^3}{8\pi} e^{-\alpha r}$$

Interpretation : If we integrate $\rho(r)$ we will find that total charge = $q_{\text{proton}} - q_{\text{electron}} = 0$
 $\rho_p(r)$ is distributed around a point, $\rho_e(r) \propto e^{-r}$ in nucleus

- 1.11 Use Gauss's theorem to prove that at the surface of a curved charged conductor, the normal derivative of the electric field is given by

$$\frac{1}{E} \frac{\partial E}{\partial n} = -\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

where R_1 and R_2 are the principal radii of curvature of the surface.



To set up the problem, we elaborate on a few points:

- * $A_1 = A$, $A_2 = A + dA$ where A_2 is the area of the surface with a change in the $\hat{n}$ direction
- * $A = \iint R_1 R_2 d\theta_1 d\theta_2$ (product of arc lengths)
- * $\frac{\partial E}{\partial n} = \lim_{\Delta r \rightarrow 0} \frac{E(\vec{x} + \Delta r \hat{n}) - E(\vec{x})}{\Delta r}$ (1)

We draw a Gaussian pillbox right outside the surface of the conductor.

$$\int \vec{E} \cdot d\vec{a} = \frac{Q}{\epsilon_0} = 0 \quad (\text{since } Q=0)$$

We also note that the sum of the flux at the top and bottom surfaces must be zero.

$$\int \vec{E}_{\text{top}} \cdot d\vec{a}' + \int \vec{E}_{\text{bottom}} \cdot d\vec{a} + \int \vec{E}_{\text{sides}} \cdot d\vec{a} = 0$$

$$E_{\text{top}} \int da' - E_{\text{bottom}} \int da = 0 \quad (\text{Electric fields } \vec{E}_{\text{top}} \text{ and } \vec{E}_{\text{bottom}} \text{ have opposite directions})$$

$$E(\vec{x} + \Delta r \hat{n}) (R_1 + \Delta r)(R_2 + \Delta r) \iint d\theta_1 d\theta_2 - E(\vec{x}) R_1 R_2 \iint d\theta_1 d\theta_2 = 0$$

$$E(\vec{x} + \Delta r \hat{n}) = \frac{E(\vec{x}) R_1 R_2}{(R_1 + \Delta r)(R_2 + \Delta r)} \quad (2)$$

Plug (2) in (1)

$$\frac{\partial E}{\partial n} = \lim_{\Delta r \rightarrow 0} \frac{\frac{E(\vec{x}) R_1 R_2}{(R_1 + \Delta r)(R_2 + \Delta r)} - E(\vec{x})}{\Delta r} = \lim_{\Delta r \rightarrow 0} \frac{\cancel{R_1 R_2} - \cancel{R_1 R_2} - \Delta r R_1 - \Delta r R_2 - \Delta r^2}{(R_1 + \Delta r)(R_2 + \Delta r) \Delta r} E(\vec{x})$$

$$= \lim_{\Delta r \rightarrow 0} - \frac{R_1 + R_2}{(R_1 + \Delta r)(R_2 + \Delta r)} E(\vec{x})$$

$$= - \frac{R_1 + R_2}{R_1 R_2} E$$

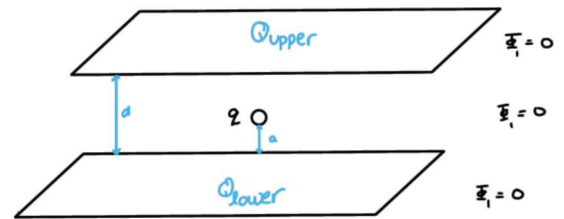
$$\Rightarrow \frac{1}{E} \frac{\partial E}{\partial n} = -\left[\frac{1}{R_1} + \frac{1}{R_2}\right]$$

- 1.13** Two infinite grounded parallel conducting planes are separated by a distance d . A point charge q is placed between the planes. Use the reciprocity theorem of Green to prove that the total induced charge on one of the planes is equal to $(-q)$ times the fractional perpendicular distance of the point charge from the other plane.
(Hint: As your comparison electrostatic problem with the same surfaces choose one whose charge densities and potential are known and simple.)

We set up 2 scenarios where

Scenario 1 : Charged plates have 0 potential.

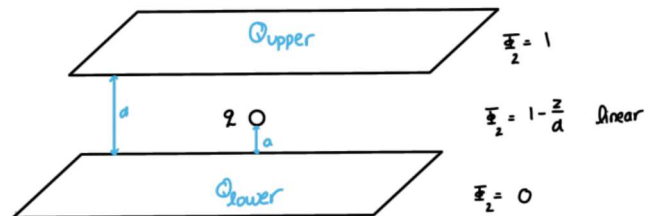
Scenario 2 : Potential difference is 1 and linear between plates



Scenario 1

We invoke Green's reciprocity Theorem :

$$\underbrace{\int_V \rho_1 \Phi_2 dV}_{I_1} + \underbrace{\int_S \sigma_1 \Phi_2 dS}_{I_2} = \underbrace{\int_V \rho_2 \Phi_1 dV}_{\cancel{I_1}} + \underbrace{\int_S \sigma_2 \Phi_1 dS}_{\cancel{I_2}}$$



Scenario 2

For I_1 , the volume only has a charge q at $z=a$.

$$\therefore I_1 = \int_V \rho_1 \Phi_2 dV = \Phi_2(a) \int_V \rho_1 dV = \Phi_2(a) q$$

For I_2 , we have 2 surfaces with S_{lower} & $\Phi_2 = 0$, $\int \sigma ds = Q_{\text{lower}}$ & S_{upper} : $\Phi_2 = 1$, $\int \sigma ds = Q_{\text{upper}}$

$$\begin{aligned} \therefore I_2 &= \int_S \sigma_1 \Phi_2 dS = \Phi_2(0) \int_{\text{lower}} \sigma ds + \Phi_2(1) \int_{\text{upper}} \sigma ds \\ &= (0)Q_{\text{lower}} + (1)Q_{\text{upper}} = Q_{\text{upper}} \end{aligned}$$

Equating I_1 & I_2 :

$$Q_{\text{upper}} = -q \Phi_2(a) = -q \left(1 - \frac{a}{d}\right)$$

$$Q_{\text{lower}} = -q - Q_{\text{upper}} = -q \left(\frac{a}{d}\right)$$

$$(Q_{\text{lower}} + Q_{\text{upper}} = -q)$$

for charge conservation.